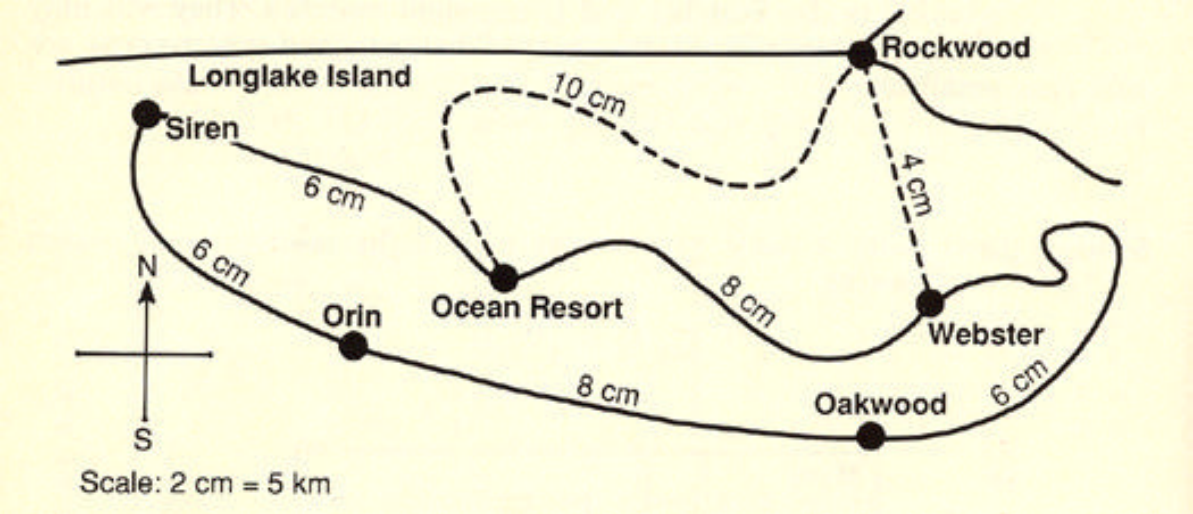
Problem: The scale on the map tells you that two centimeters on the map represents an actual distance of five kilometers. How is the actual distance between towns determined from the information on the map?



Solution plan: Build a table of data to show the actual distance *D* in kilometers represented by a map distance *d* measured in centimeters.

Map distance (d)	2 cm	4 cm	6 cm	8 cm	10 cm
Actual distance (D)					

Questions:

1. Use the table to determine the actual distances (shortest by road) between the following:
   Rockwood and Webster _____ Ocean Resort and Webster _____
   Rockwood and Ocean Resort _____ Siren and Ocean Resort _____

2. What actual distance is represented by 1 cm on the map?

3. Write a formula that can be used to determine the actual distance *D* between two towns if you know the map distance d between the towns.
   $D =$ _____

4. Use the formula to find the actual distance from Webster to Siren.

5. Graph the data from the table on a pair of coordinate axes.

6. Does this problem represent a proportional relationship?Support your answer using characteristics of both the formula and the graph of the data from this situation.

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